Positive Moment Bounds and Growing Plunge Windows

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Abstract

The reciprocal trace associated with the even half-filled sine kernel has a positive Stieltjes representation. It follows that every finite inverse-power expansion has a remainder of the sign of, and smaller than, its first omitted term. Two rational bounds control the exact central slope. The full formal inverse-power series nevertheless has zero radius of convergence. An elementary variational construction controls growing central windows of the commuting Jacobi matrix. An exact Darboux deletion, with its singular endpoint domains proved explicitly, realizes the omitted spectrum as a positive differential operator. Elementary bracketing then supplies higher reciprocal moments and proves the signed plunge law uniformly for $|\lambda| \le L_N = o(N^{1/4})$. In particular a logarithmic number of actual central pairs obey the law with relative error $O(N^{-1/2})$. No bulk quantum-period remainder is assumed.

1 The finite problem and exact inputs

Let $N = 2m$ be a positive even integer. With site indices $1, \dots, N$, define

$$B_{jk} = \begin{cases} 1/2, & j = k, \\ \sin[\pi(j-k)/2]/[\pi(j-k)], & j \neq k, \end{cases} \qquad (J_N)_{j,j+1} = (J_N)_{j+1,j} = b_j := \frac{j(N-j)}{N}, \quad (J_N)_{jj} = 0.$$

The matrices commute and $0 < B < I$. Write their common eigenvalues as $J_N v = \lambda v$ and $Bv = \zeta v$. Let $H = NJ_N/2$, $\alpha = (N^2-1)/4$, and let $\mu_{N,k}$, in increasing order starting at $k = 0$, be the eigenvalues of the Friedrichs realization

$$A_N u = -(\cos x\, u')' - \alpha \cos x\, u, \qquad -\pi/2 < x < \pi/2.$$

The endpoint condition excludes the logarithmic solution. The exact spectral-tail and trace identities established in [1, 2] are

$$\{\mu_{N,0}, \dots, \mu_{N,N-1}\} = -\operatorname{spec} H, \qquad \mu_{N,N} > \|H\|, \qquad Z_N := \sum_{k \ge N} \mu_{N,k}^{-1}, \tag{1}$$

$$\log \frac{\zeta}{1-\zeta} = F_N(N\lambda/2), \qquad F_N(b) = 2 \sum_{k \ge N} \operatorname{artanh} \frac{b}{\mu_{N,k}}, \tag{2}$$

$$Z_N = \pi \int_0^\infty e^{-Nt} f(t) \mathrm{d}t, \qquad f(t) = P_{-1/2}(\cosh t)^2. \tag{3}$$

Here P is the real Legendre function on $[1, \infty)$. The first identity identifies precisely the modes removed from the differential spectrum; the second holds at the actual common finite modes. The normalization in (3) follows from a weighted Legendre resolvent and a Mellin reflection identity, rather than an asymptotic fit. These previous results are the exact inputs to this paper. The positive integral in (3) defines an interpolation for every real $N > 0$; equality with the specified spectral trace is used only for even integers.

2 A positive Fourier measure

Lemma 1 (Autocorrelation identity)**.** *Put* $g(u) = \operatorname{sech}(u)^{1/2}$ *and use the Fourier convention* $\widehat{g}(\omega) = \int_{\mathbb{R}} e^{-i\omega u} g(u)\mathrm{d}u$*. Then*

$$y(t) := P_{-1/2}(\cosh t) = \frac{1}{\pi}\int_{\mathbb{R}} g(u)g(u-t)\mathrm{d}u, \tag{4}$$

$$\widehat{g}(\omega) = \frac{|\Gamma(1/4+i\omega/2)|^2}{\sqrt{2\pi}}, \qquad y(t) = \int_{\mathbb{R}} e^{i\omega t} p(\omega)\mathrm{d}\omega, \tag{5}$$

$$p(\omega) = \frac{|\Gamma(1/4+i\omega/2)|^4}{4\pi^3}. \tag{6}$$

The function p *is an even, strictly positive probability density with moments of every order. Consequently* $f = y^2$ *is the Fourier transform of the probability density* $q = p * p$*.*

Proof. The classical angular representation [3, Eq. 14.12.7] gives

$$y(t) = \frac{1}{\pi}\int_0^{\pi} (\cosh t + \sinh t \cos\theta)^{-1/2}\mathrm{d}\theta.$$

For $t \geq 0$, substitute $\tan(\theta/2) = e^u$. Then $\mathrm{d}\theta = \operatorname{sech} u\,\mathrm{d}u$ and

$$\cosh t + \sinh t\cos\theta = \frac{\cosh(u-t)}{\cosh u},$$

which proves (4). Both sides are even in t, so it holds on the whole real line. Substitution $x = e^{2u}$ in the Fourier integral, followed by Euler's beta integral [3, Eq. 5.12.3], gives

$$\widehat{g}(\omega) = 2^{-1/2}\int_0^{\infty} x^{1/4-i\omega/2-1}(1+x)^{-1/2}\mathrm{d}x = \frac{\Gamma(1/4-i\omega/2)\Gamma(1/4+i\omega/2)}{\sqrt{2\pi}}.$$

The function g and all its derivatives decay exponentially; in particular g is a Schwartz function. Fourier inversion for its autocorrelation proves (5). Plancherel and $\int_{\mathbb{R}} g^2\mathrm{d}u = \pi$ give $\int p = 1$. Positivity follows because the gamma function has no zeros. The Schwartz property supplies every moment and justifies differentiation under the Fourier integral. Multiplication of the two absolutely convergent Fourier integrals gives the convolution statement. □

Let σ be the image of $q(\omega)\mathrm{d}\omega$ under $s = \omega^2$. It is a probability measure on $[0,\infty)$, has full support, and has moments $m_k = \int_0^{\infty} s^k \mathrm{d}\sigma(s)$ for all $k \geq 0$.

Theorem 2 (Stieltjes representation and signed remainder)**.** *For the interpolation in* (3)*, for every* $N > 0$*,*

$$\boxed{\frac{NZ_N}{\pi} = \int_0^{\infty} \frac{\mathrm{d}\sigma(s)}{1+s/N^2}.} \tag{7}$$

For every integer $K \geq 0$*,*

$$0 < (-1)^{K+1}\left[\frac{NZ_N}{\pi} - \sum_{k=0}^{K}\frac{(-1)^k m_k}{N^{2k}}\right] < \frac{m_{K+1}}{N^{2K+2}}. \tag{8}$$

The formal series in N^{-2} *has zero radius of convergence, while* (8) *proves its asymptotic validity to every fixed order.*

Proof. Insert the Fourier representation for f in (3). Absolute Fubini is justified by $\int_{\mathbb{R}} q(\omega)\int_0^{\infty} Ne^{-Nt}\mathrm{d}t\,\mathrm{d}\omega = 1$. Integrating in t and using the evenness of q gives $N^2/(N^2+\omega^2)$, proving (7). The finite identity

$$\frac{1}{1+x} = \sum_{k=0}^{K}(-x)^k + \frac{(-1)^{K+1}x^{K+1}}{1+x}, \qquad x \geq 0,$$

then gives the signed remainder in (8) exactly as

$$\frac{1}{N^{2K+2}}\int_0^{\infty}\frac{s^{K+1}}{1+s/N^2}\mathrm{d}\sigma(s).$$

It is strictly between zero and m_{K+1}/N^{2K+2} because σ is not supported at zero. For every $R > 0$, $m_k \geq R^k \sigma([R, \infty))$, with a strictly positive last factor. Therefore $m_k^{1/k} \to \infty$, proving zero convergence radius. The estimate at each fixed K already proves the asymptotic assertion. □

Since $f^{(2k)}(0) = (-1)^k m_k$, its differential equation or hypergeometric series determines the moments exactly:

$$\begin{array}{c|cccccc} k & 0 & 1 & 2 & 3 & 4 & 5 \\ \hline m_k & 1 & \frac{1}{4} & \frac{17}{64} & \frac{169}{256} & \frac{50777}{16384} & \frac{1575169}{65536}. \end{array} \tag{9}$$

For example, if

$$S_N = \frac{\pi}{N} - \frac{\pi}{4N^3} + \frac{17\pi}{64N^5} - \frac{169\pi}{256N^7} + \frac{50777\pi}{16384N^9},$$

then for every positive even N,

$$0 < S_N - Z_N < \frac{1575169\pi}{65536N^{11}}. \tag{10}$$

These are bounds on the exact trace, not just on formal coefficients.

Corollary 3 (Two rational slope bounds)**.** *For every positive even N,*

$$\boxed{\frac{4\pi}{16N^2 + 17} < \pi - NZ_N < \frac{\pi}{4N^2 + 1}.} \tag{11}$$

Also

$$\left|\pi - NZ_N - \frac{\pi}{4N^2}\right| < \frac{17\pi}{64N^4}. \tag{12}$$

Proof. Strict Jensen applied to $s \mapsto (1 + s/N^2)^{-1}$ under σ gives $NZ_N/\pi > (1 + m_1/N^2)^{-1}$. For the probability measure $\mathrm{d}\tau = s\mathrm{d}\sigma/m_1$, its mean is $m_2/m_1 = 17/16$ and

$$1 - \frac{NZ_N}{\pi} = \frac{m_1}{N^2} \int \frac{\mathrm{d}\tau(s)}{1 + s/N^2} > \frac{m_1}{N^2 + m_2/m_1}.$$

Both measures are nondegenerate, so both inequalities are strict. Substitution of the moments proves (11); the case $K = 1$ of (8) proves (12). □

3 A growing window of small Jacobi eigenvalues

List the positive eigenvalues of J_{2m} increasingly as $0 < \lambda_{1,+} \leq \cdots \leq \lambda_{m,+}$. The negative eigenvalues are their opposites: conjugation by alternating signs sends J to $-J$. There is no zero eigenvalue, since $\det J_{2m} = (-1)^m \prod_{k=1}^m b_{2k-1}^2 \neq 0$.

Theorem 4 (Explicit logarithmic-window bound)**.** *Set $L = \lfloor m/2 \rfloor$. For any integer $r \geq 1$ with $16^r \leq L$, put $\ell = \lfloor \log L/(r \log 16) \rfloor$. Then*

$$\lambda_{r,+} \leq \frac{12\sqrt{2}\,\pi}{\ell \log 16}. \tag{13}$$

In particular, if $1 \leq r \leq \log L/(2 \log 16)$, then

$$\boxed{\lambda_{r,+} \leq \frac{24\sqrt{2}\,\pi r}{\log L}.} \tag{14}$$

Thus the innermost r_N pairs shrink uniformly to zero whenever $r_N = o(\log N)$ as even $N \to \infty$.

Proof. Define $A_1 = 1$ and $A_{k+1} = -b_{2k-1}A_k/b_{2k}$ for $k < m$. For $v_{2k-1} = A_k f_k$, $v_{2k} = 0$, direct substitution gives

$$\|v\|^2 = \sum_{k=1}^m A_k^2 f_k^2, \qquad \|Jv\|^2 = \sum_{k=1}^m (b_{2k-1}A_k)^2 (f_k - f_{k+1})^2, \tag{15}$$

where $f_{m+1} = 0$ encodes the actual last row. For $1 \le k \le L$,

$$\frac{1}{4k} \le A_k^2 \le \frac{3}{k}, \qquad (b_{2k-1}A_k)^2 \le 12k. \tag{16}$$

Indeed, with $c_s = (1/2)_s/s!$,

$$|A_k| = c_{k-1} \prod_{j=1}^{k-1} \left(1 + \frac{1}{2(m-j)}\right).$$

For $s \ge 1$, sc_s^2 increases from $1/4$, because its successive ratio is $(2s+1)^2/[4s(s+1)] > 1$. Also $(s+1)c_s^2$ decreases from one, because its successive ratio is $1 - (3s+2)/[4(s+1)^3] < 1$. These prove the required lower bound and $c_{k-1}^2 \le 1/k$. For $k \le L$, the logarithm of the squared product is at most

$$\sum_{j=1}^{k-1} \frac{1}{m-j} \le \frac{k-1}{m-k+1} < 1.$$

The squared product is therefore below $e < 3$. Checking $k = 1$ directly and using $b_{2k-1} \le 2k$ proves (16).

Set $R = 16^\ell$ and $h = \log R$. On each integer block $[a, b] = [R^j, R^{j+1}]$, $j = 0, \dots, r-1$, use the sequence

$$f_k = \begin{cases} \sin[\pi \log(k/a)/h], & a \le k \le b, \\ 0, & \text{otherwise.} \end{cases}$$

All blocks lie in $[1, L]$, and their shared endpoints are zero. The resulting odd-site vectors have disjoint nonzero supports; so do their images under J, since a block contributes only to the even rows corresponding to $k = a, \dots, b-1$ in (15). The norm and energy forms consequently have no cross terms between different packets.

The numbers $aR^{1/4}$ and $aR^{3/4}$ are integers, and $f_k^2 \ge 1/2$ between them. Thus

$$\|v\|^2 \ge \frac{1}{8} \sum_{k=aR^{1/4}}^{aR^{3/4}} \frac{1}{k} \ge \frac{1}{8} \int_{aR^{1/4}}^{aR^{3/4}+1} \frac{\mathrm{d}x}{x} \ge \frac{h}{16}.$$

The sine function is 1-Lipschitz and $\log(1 + 1/k) \le 1/k$, so $|f_{k+1} - f_k| \le \pi/(hk)$ inside the block. Therefore

$$\|Jv\|^2 \le \frac{12\pi^2}{h^2} \sum_{k=a}^{b-1} \frac{1}{k} \le \frac{12\pi^2(h+1)}{h^2} \le \frac{18\pi^2}{h},$$

where $h \ge \log 16 > 2$. Every nonzero vector in the r-dimensional packet span thus has $\|Jv\|^2/\|v\|^2 \le 288\pi^2/h^2$.

Order the odd sites before the even sites. Then

$$J = \begin{pmatrix} 0 & T^{\mathsf{T}} \\ T & 0 \end{pmatrix},$$

where T is invertible and upper bidiagonal, with diagonal $b_1, b_3, \dots, b_{2m-1}$. Its singular values are precisely the positive eigenvalues of J. Min–max on the odd-site block $T^{\mathsf{T}}T$ therefore gives $\lambda_{r,+}^2 \le 288\pi^2/h^2$, proving (13). If $\log L/(r \log 16) \ge 2$, the floor is at least half its argument; this proves (14). Finally $\log L \sim \log N$. □

The constants are deliberately elementary, not optimal. The finite-size condition for (13) already requires $N \ge 64$ when $r = 1$; the assertion is not applied outside its stated range.

4 Uniform transfer to the modular error

For a common mode define

$$\eta(t) = \pi t \, {}_3F_2(1/4, 1/2, 3/4; 1, 3/2; 4t^2), \qquad \varepsilon = \log \frac{1-\zeta}{\zeta}, \qquad e = \varepsilon + N\eta(\lambda/N).$$

All quantities are dimensionless. In particular the word “modular” here does not supply a physical clock or an energy unit.

Lemma 5 (Explicit ratio control)**.** *For every even $N \geq 8$ and every common mode with $0 < |\lambda| \leq 1$,*

$$\left| \frac{-4N^2 e}{\varepsilon} - 1 \right| \leq \frac{10}{N^2} + 33\lambda^2. \tag{17}$$

Proof. The two central sites give $\|H\| \geq N^2/8$ by a Rayleigh quotient. From (1), with $\beta = N\lambda/2$,

$$\frac{|\beta|}{\mu_{N,N}} \leq \frac{4|\lambda|}{N} \leq \frac{1}{2}, \qquad \sum_{k \geq N} \mu_{N,k}^{-3} \leq \frac{64 Z_N}{N^4} \leq \frac{64\pi}{N^5}.$$

The positive artanh series therefore yields

$$|F_N(\beta) - N\lambda Z_N| \leq \frac{64\pi|\lambda|^3}{9N^2}. \tag{18}$$

The coefficients d_k of the hypergeometric series in the definition of η are positive and decreasing, with $d_1 = 1/16$. Indeed their successive ratio is $[(k+1/4)(k+1/2)(k+3/4)]/[(k+1)^2(k+3/2)] < 1$. Thus

$$|N\eta(\lambda/N) - \pi\lambda| \leq \frac{\pi|\lambda|^3}{4N^2(1 - 4\lambda^2/N^2)} \leq \frac{4\pi|\lambda|^3}{15N^2}.$$

Since $\varepsilon = -F_N(\beta)$, combining this with (18) gives

$$|e - \lambda(\pi - NZ_N)| \leq \frac{8\pi|\lambda|^3}{N^2}.$$

Set $a = 4N^2 e/(\pi\lambda)$ and $b = F_N(\beta)/(\pi\lambda)$. Equations (12) and (18), including the common sign of each artanh term and λ, imply

$$|a - 1| \leq \frac{17}{16N^2} + 32\lambda^2, \qquad b \geq 1 - \frac{1}{4N^2} > 0, \qquad |b - 1| \leq \frac{8}{N^2}.$$

Consequently

$$\left| \frac{a}{b} - 1 \right| \leq \frac{145/(16N^2) + 32\lambda^2}{1 - 1/(4N^2)} \leq \frac{10}{N^2} + 33\lambda^2.$$

Here $a/b = -4N^2 e/\varepsilon$. Strict positivity of b also justifies division by ε. □

Corollary 6 (Uniform signed plunge law on a growing window)**.** *Let $r_N \geq 1$ be integers with $r_N = o(\log N)$ as even $N \to \infty$. For both signs of the innermost r_N pairs of common eigenmodes,*

$$\sup_{1 \leq j \leq r_N,\, \pm} \left| \frac{-4N^2 e_{j,\pm}}{\varepsilon_{j,\pm}} - 1 \right| = O\left(N^{-2} + \frac{r_N^2}{\log^2 N} \right) \longrightarrow 0. \tag{19}$$

For all sufficiently large N, an explicit upper bound for the left side is

$$\frac{10}{N^2} + 33 \left(\frac{24\sqrt{2}\,\pi r_N}{\log\lfloor N/4 \rfloor} \right)^2.$$

Proof. The bound (14) puts every eigenvalue in the window below one in absolute value for all sufficiently large N. Apply (17) to each mode. Its constants are uniform, so taking the supremum preserves the estimate. □

In particular every fixed number of central pairs obeys the law, with the conservative relative error $O((\log N)^{-2})$.

5 An exact positive operator for the differential tail

The higher reciprocal moments admit a useful positive realization. Write $p(x) = \cos x$ and introduce the Liouville coordinate

$$z(x) = \int_{-\pi/2}^{x} \frac{\mathrm{d}u}{\sqrt{\cos u}}, \qquad 0 < z < \mathcal{L}, \qquad \mathcal{L} = \frac{\sqrt{\pi}\,\Gamma(1/4)}{\Gamma(3/4)}.$$

Below $p(z) = \cos x(z)$.

Theorem 7 (Exact deletion of the finite block)**.** *The spectrum of*

$$\boxed{T_N = -D_z^2 + \frac{4N^2-1}{16p(z)} + \frac{p(z)}{16}} \tag{20}$$

with its Friedrichs realization is precisely $\{\mu_{N,k} : k \geq N\}$*. Its form domain is* $H_0^1(0,\mathcal{L})$*, and its endpoints are limit point. In particular,*

$$\mu_{N,N} > N^2/4, \qquad Z_N(s) := \sum_{k\geq N} \mu_{N,k}^{-s} = \operatorname{Tr} T_N^{-s} \quad (s > 1/2). \tag{21}$$

Proof. The unitary map $\psi(z) = p(x(z))^{1/4}u(x(z))$ transforms A_N into $-D_z^2 + V_N$, where direct differentiation gives

$$V_N = -N^2 p/4 + p/16 - 1/(16p).$$

At either endpoint, with d the distance to that endpoint, $p = d^2/4 + O(d^6)$. Thus $V_N = -1/(4d^2) + O(d^2)$, and the selected eigenfunctions behave as $d^{1/2}$ times a regular power series in d^2. The logarithmic branch is excluded. This critical endpoint requires a domain argument in addition to the usual formal Darboux–Crum transformation [4].

For one deletion, let $g > 0$ be the current ground eigenfunction, with eigenvalue E_0, and suppose its selected endpoint exponent is $k + 1/2$, $k \geq 0$. Define Q as the closure from C_c^∞ of $D_z - g'/g$. The compact-support ground-state identity, followed by closure of quadratic forms, gives

$$S - E_0 = Q^*Q$$

for the current Friedrichs operator S. The self-adjoint partner $QQ^* + E_0$ extends the differential expression with potential $V - 2(\log g)''$. Differentiable Frobenius expansions show that its endpoint coefficient is $((k+1)^2 - 1/4)d^{-2} + O(1)$. Its two branches have exponents $k + 3/2$ and $-k - 1/2$; only the first is square integrable. Hence the partner is limit point at both endpoints and is its unique self-adjoint realization. Moreover $\ker Q^* = \{0\}$, since $Q^*f = 0$ implies $f = c/g$, which is not square integrable, including at $k = 0$. The gap above E_0 makes $\operatorname{ran} Q$ closed; its orthogonal complement is $\ker Q^*$, so it is all of L^2. Polar decomposition therefore pairs the orthogonal complement of g unitarily with the partner space. Exactly the ground eigenvalue is deleted, with completeness and compact resolvent preserved. The new ground state has exponent $k + 3/2$, permitting iteration.

Direct finite Fourier transformation and (1) show that the first N original eigenfunctions span

$$\operatorname{span}\{e^{imx} : m = -(N-1)/2, \ldots, (N-1)/2\}.$$

Their transformed Wronskian is therefore

$$W_z(\psi_0, \ldots, \psi_{N-1}) = C\, p^{N^2/4}, \qquad C \neq 0.$$

Indeed the common factor $p^{1/4}$ contributes $p^{N/4}$, the change of variable contributes $(\mathrm{d}x/\mathrm{d}z)^{N(N-1)/2} = p^{N(N-1)/4}$, and the Fourier Wronskian is a nonzero constant Vandermonde determinant. Iterating the one-step identity gives $V_{\text{tail}} = V_N - 2D_z^2 \log|W_z|$. Since

$$D_z^2 \log p = -\tfrac{1}{2}(p + p^{-1}),$$

this is exactly (20). The Wronskian formula evaluates the already justified self-adjoint deletions; it does not replace their domain argument.

The final potential is $(N^2 - 1/4)d^{-2} + O(1)$. Hardy's inequality gives form domain H_0^1. Finally

$$\frac{4N^2-1}{16p} + \frac{p}{16} - \frac{N^2}{4} = \frac{(1-p)(4N^2-1-p)}{16p} \geq 0.$$

The kinetic energy is strictly positive on H_0^1, proving the first assertion of (21). The trace convergence and its quantitative form also follow from the bracketing below. □

6 Higher reciprocal moments by elementary bracketing

Theorem 8 (Higher-moment leading terms with an explicit error)**.** *For each fixed* $s > 1/2$*,*

$$Z_N(s) = N^{1-2s}\{I_s + O_s(N^{-1/2})\}, \qquad I_s = \frac{2^{2s-2}\Gamma(s-1/2)\Gamma(s/2)}{\Gamma(s)\Gamma((s+1)/2)}. \tag{22}$$

For every integer $M \geq 1$ one has, in particular,

$$\left|N^5 Z_N(3) - 3\pi\right| \leq D_{N,M} := \frac{6\mathcal{L}}{M} + \frac{128M}{N} + 3\pi\left[\left(1 - \frac{1}{4N^2}\right)^{-5/2} - 1\right]. \tag{23}$$

Proof. Put $h = 1/N$ and $Q_h = h^2 T_N = -h^2 D_z^2 + V_h$. Then

$$V_h = \left(1 - \frac{h^2}{4}\right) V_0 + \frac{h^2 p}{16}, \qquad V_0 = \frac{1}{4p}, \qquad (1 - h^2/4)V_0 \leq V_h \leq V_0.$$

The potential V_h has minimum $1/4$ and is monotone on each half-interval. Divide each half into M cells of length $\delta = \mathcal{L}/(2M)$. Neumann conditions and the cell infimum of V_h give an upper trace bound by the min-max principle. Dirichlet conditions and the cell supremum give a lower trace bound. The two exterior cells, whose supremum is infinite, are omitted from the latter bound; this is a restriction of variational subspaces, not a boundedness assumption at the singular endpoints.

For constant potential $v > 0$ the cell eigenvalues are $(h\pi j/\delta)^2 + v$, with $j \geq 0$ for Neumann and $j \geq 1$ for Dirichlet conditions. The integral test gives

$$\left|h \operatorname{Tr}(-h^2 D_{\mathrm{N/D}}^2 + v)^{-s} - c_s \delta v^{1/2-s}\right| \leq h v^{-s}, \qquad c_s = \frac{\Gamma(s - 1/2)}{2\sqrt{\pi}\,\Gamma(s)}.$$

The total cell trace error is at most $2Mh4^s$. The function $V_h^{1/2-s}$ rises from zero to $4^{s-1/2}$ and falls symmetrically; its upper and lower Riemann sums differ by at most $2\delta 4^{s-1/2}$. Therefore, with $I_s(h) = c_s \int_0^{\mathcal{L}} V_h^{1/2-s}\mathrm{d}z$,

$$\left|h \operatorname{Tr} Q_h^{-s} - I_s(h)\right| \leq \frac{c_s \mathcal{L} 4^{s-1/2}}{M} + 2Mh4^s.$$

This also proves convergence of the trace. Potential comparison gives $I_s \leq I_s(h) \leq (1 - h^2/4)^{-(s-1/2)} I_s$, where

$$I_s = c_s 4^{s-1/2} \int_{-\pi/2}^{\pi/2} (\cos x)^{s-1} \mathrm{d}x.$$

The beta integral evaluates this as (22). Since $h \operatorname{Tr} Q_h^{-s} = N^{2s-1} Z_N(s)$, choosing $M = \lceil \sqrt{N} \rceil$ proves the stated rate. For $s = 3$, $c_3 = 3/16$ and $I_3 = 3\pi$, giving (23). □

For example $I_5 = 105\pi/4$. The first-moment result obtained by this general argument is weaker than (3); the new information is the independently proved higher-moment coefficient. No bulk period expansion was used.

7 A larger uniform eigenvalue window

The third-moment coefficient cancels the leading nonlinear term of the modular error.

Theorem 9 (Uniform center law beyond shrinking eigenvalues)**.** *Let $L_N = o(N^{1/4})$ as even $N \to \infty$. Uniformly over nonzero common modes with $|\lambda| \leq L_N$,*

$$e = \frac{\pi\lambda}{4N^2}\left[1 + O\left(N^{-2} + \lambda^2 N^{-1/2} + \lambda^4 N^{-2}\right)\right], \tag{24}$$

$$\frac{-4N^2 e}{\varepsilon} \longrightarrow 1. \tag{25}$$

This includes every fixed bounded eigenvalue window. More explicitly, for even $N \geq 2$, $0 < |\lambda| \leq N/4$, and any integer $M \geq 1$,

$$\left|\frac{-4N^2 e}{\varepsilon} - 1\right| \leq \frac{\dfrac{21}{16N^2} + \left(\dfrac{D_{N,M}}{3\pi} + \dfrac{16}{9N^2}\right)\lambda^2 + \dfrac{20\lambda^4}{N^2}}{1 - \dfrac{1}{4N^2}}. \tag{26}$$

Proof. Set $\beta = N\lambda/2$. By (21), $r = |\beta|/\mu_{N,N} < 2|\lambda|/N \le 1/2$. The exact artanh series (2) gives

$$F_N(\beta) = N\lambda Z_N + \frac{N^3\lambda^3}{12} Z_N(3) + R, \qquad |R| \le \frac{2|\beta|^5}{5(1-r^2)} Z_N(5).$$

The simple bound $Z_N(5) \le (4/N^2)^4 Z_N \le 256\pi/N^9$ suffices, giving $|R| \le 64\pi|\lambda|^5/(15N^4)$. The convergent hypergeometric series defining η, with positive decreasing coefficients, gives

$$N\eta(\lambda/N) = \pi\lambda + \frac{\pi\lambda^3}{4N^2} + R_\eta, \qquad |R_\eta| \le \frac{7\pi|\lambda|^5}{16N^4}.$$

Equation (23) cancels the two cubic terms, and (12) therefore implies

$$\left| e - \frac{\pi\lambda}{4N^2} \right| \le \frac{17\pi|\lambda|}{64N^4} + \frac{D_{N,M}|\lambda|^3}{12N^2} + \frac{5\pi|\lambda|^5}{N^4}.$$

Choosing $M = \lceil\sqrt{N}\rceil$ proves (24). For division, put $a = 4N^2 e/(\pi\lambda)$ and $b = F_N(\beta)/(\pi\lambda)$. The sign of every artanh term gives $b \ge NZ_N/\pi \ge 1 - 1/(4N^2) > 0$. Also $Z_N(3) \le 16Z_N/N^4$ and the artanh bound before the cubic term give

$$|b-1| \le \frac{1/4 + 16\lambda^2/9}{N^2}.$$

Combining this with the preceding bound on $a-1$ gives (26), since $a/b = -4N^2 e/\varepsilon$. On the stated window all its terms tend uniformly to zero, proving (25). □

Corollary 10 (A logarithmic number of actual central pairs)**.** *For all sufficiently large even N, put*

$$r_N = \left\lfloor \frac{\log\lfloor N/4\rfloor}{\log 16} \right\rfloor.$$

For both signs of the innermost r_N pairs of common modes,

$$\sup_{1\le j\le r_N,\,\pm} \left| \frac{-4N^2 e_{j,\pm}}{\varepsilon_{j,\pm}} - 1 \right| = O(N^{-1/2}).$$

Proof. The choice of r_N satisfies $16^{r_N} \le \lfloor N/4\rfloor$. The integer ℓ in (13) is at least one, so

$$|\lambda_{j,\pm}| \le \frac{12\sqrt{2}\,\pi}{\log 16} \qquad (1 \le j \le r_N).$$

Apply (26) on this fixed bounded window with $M = \lceil\sqrt{N}\rceil$. Its constants are independent of the selected mode. The numerical constant and the bracketing rate are not asserted to be optimal. □

This is still a center result: $t = \lambda/N = o(N^{-3/4})$. The argument proves no fixed-t bulk quantum-period remainder and does not claim the displayed bracketing rate or window exponent is optimal.

8 Literature comparison and scope

The logarithmic scale has a rigorous precedent. Deift, Its, and Krasovsky [5, Theorem 1.3] prove that, for a symbol taking two positive constant values on complementary rational arcs, eigenvalues in a fixed compact subinterval of the gap have spacing of order $1/\log N$; every interval of a suitable constant times $1/\log N$ contains an eigenvalue. The choice of values $1, 2$ and a half-circle gives $I + B_N$ up to diagonal unitary conjugation. This supports the known logarithmic occupation scale. The finite Jacobi construction above supplies an independent bound with explicit constants. No new scale or priority is claimed.

The positive-measure theorem controls the exact first reciprocal trace at every even size. The growing-window result concerns actual common eigenmodes, and its proof is uniform over the stated window. It does not establish a uniform bulk quantum-period expansion, an odd-size deflation identity, or a general-filling effective-size convention. The fixed-interior bulk expansion is supplied separately by *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law*; it is not a consequence of moment bracketing alone. In particular, the trace measure's

moments m_k are not the higher reciprocal spectral traces $\sum_{j\geq N} \mu_{N,j}^{-(2k+1)}$. Their leading asymptotics are established separately by (22).

Exact arithmetic checks recover the displayed moments independently from the hypergeometric series. Quadrature checks compare the autocorrelation, gamma density, and Legendre representations, and finite matrix checks verify the packet energies. Symbolic checks also verify the transformed potential, Wronskian exponent, and moment constants. These checks provide reproducibility; the analytic inequalities and min–max argument are the proofs. No finite numerical agreement is used to infer a theorem.

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References

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